03 Newtons Law

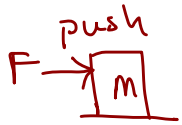
Friday, November 10, 2023 2:39 AM

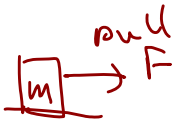
Ch 4 & Ch 5

Newton's Law — Applications of Newton's Law.

x, v, a ✓

For CF → most objects



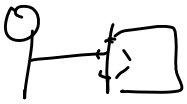


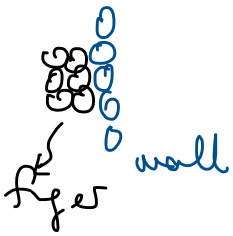
} daily experience 'contact' ≡ touch

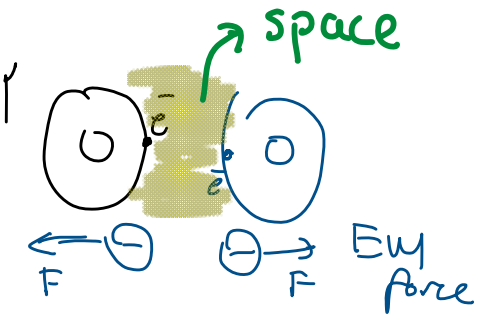
For RCF (4)

- → grav. force ⇒ phys 1
- → Electromagnetic (EM) ⇒ phys 1 & phys 2
- → Nuclear strong / weak force } QM Nuclear physics

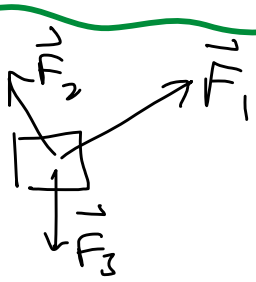
contact forces are EM force.



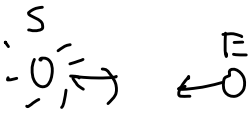




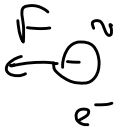
Force = vector
= magnitude
direction



$F = [\text{Newton}] = [N]$



$F \sim 10^{22} N$



$F \sim 10^{-30} N$?

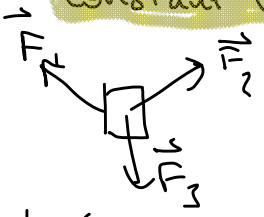
scale = test, scale = 80 kg F = weight

Newton's unit

Newton's LAW

1st (inertia)

If there's no net force acts on an object, object moves with constant velocity



$\sum \vec{F} = 0$

$v = \text{const.}$

net force = total force

const velocity ≡ it keeps its state

inertia ≡ eyelensight

"tendency to keep its state"

↑inertia ~ mass ↑

relative motion

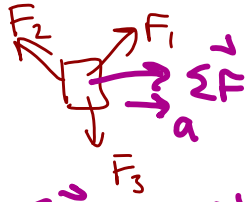
$\vec{u}_{P/A} = \vec{a}_{P/B} + \vec{a}_{B/A}$

$\vec{a}_{B/A} \to 0$

A & B are "inertial frame of reference"

2nd

If a net force acts on an object object has an acceleration



$\sum \vec{F} = m\vec{a}$

$[N] = [kg \frac{m}{s^2}]$

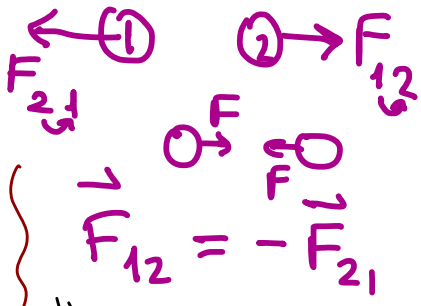
m = mass = amount of inertia

$m = \frac{\sum \vec{F}}{\vec{a}} = \frac{F}{g} = \frac{F}{a}$

$\sum \vec{F}$ = net force
total

3rd

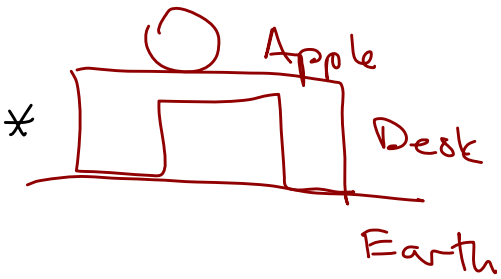
Between two objects if 1st one exerts a force onto 2nd one, then 2nd one exerts the same force onto 1st.



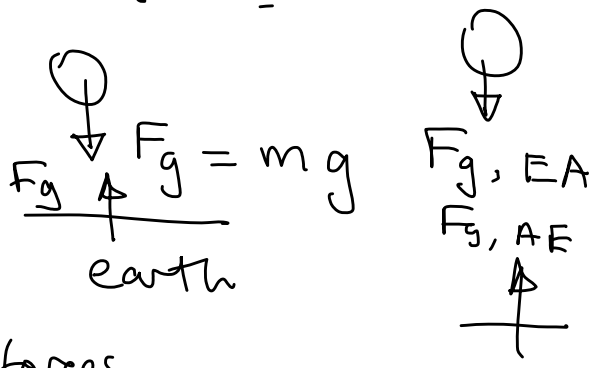
$\vec{F}_{12} = -\vec{F}_{21}$

"action-reaction pairs of forces"

"action reaction forces act on different objects"



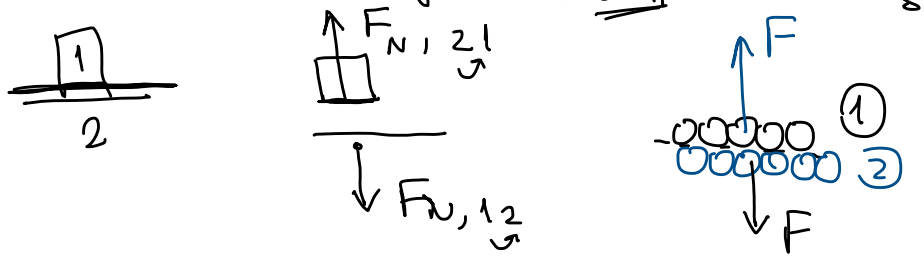
contact force ≡



contact forces ≡ forces due to , forces due

contact forces ≡ forces due to surfaces , forces due to strings

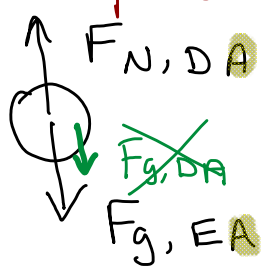
surfaces PUSH STRING PULLS



FREE BODY DIAGRAM: show all forces acting on an object.
(FBD)

forces acting on Apple ① ②

FBD

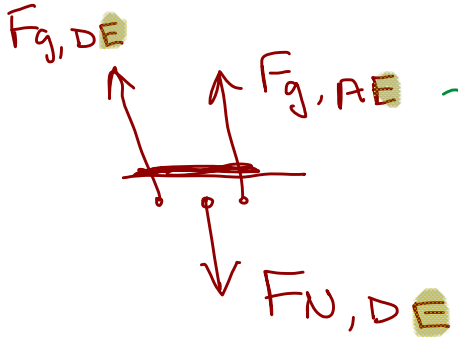


$F_g = G \frac{M_1 M_2}{r^2}$

2 surface
2 contact forces

FBD

Mass of object smaller than Mass of earth



FBD

FBD ⇒ $\Sigma \vec{F} = m\vec{a}$ $a = ?$

$\Sigma \vec{F} = 0$
$\vec{a} = 0$